Dealerless Forward-FC Preprocessing for Orca: GPU Ring-LPN Design, Proof Boundary, and Evaluation (v2.5)

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2026-08-04 (v2.5; live two-process composition and model-scale evidence)

Abstract

Orca's GPU secure-inference/training system uses trusted-dealer preprocessing for its linear layers. We remove that dealer from one complete forward fully-connected (FC) preprocessing path: two independent processes sample party-local masks and sparse Ring-LPN noise, jointly sample the full public polynomial, generate standard distributed point-function keys over SCI/IKNP and Gilboa OLE, expand Figure-2 Ring-LPN OLEs on separate GPUs, convert exact CRT shares to $\mathbb{Z}_{2^{32}}$, and emit byte-compatible Orca keys. Neither live process reads peer-private state; an omniscient checker runs only after both exit and validates the keys through Orca's unchanged `gpuMatmulBeaver`.

The proof contribution is explicit and scoped. A level-by-level coupling shows that distributed key generation emits exactly the standard DPF correction words conditioned on the two roots; role-specific simulators cover correlated batches and the three-OLE payload step without the previously leaked control sign. A separate simulator proves the exact share conversion private in the edaBit/daBit/Boolean-triple hybrid. Under standard DPF/PRG and OT/OLE security, authenticated channels, and decisional Ring-LPN for the exact sparse distribution, these pieces give a conditional static-semi-honest theorem for forward FC.

At the exact ResNet18 classifier-layer shape $1\times512\times1000$, ten measured local-loopback trials all pass. Median preprocessing is 35.735 s and sends 608,860,824 application bytes, versus 10.813 ms for matched stock trusted-dealer key generation; final payload size remains 4,108,096 bytes per party and the unchanged two-share online checker takes 1.099 ms. This is a forward-FC preprocessing/checker measurement, not full-model inference or truncation; the negative comparison establishes feasibility and the optimization target, not a speedup. The exercised $(n, c, t) = (8192, 2, 8)$ setting is a correctness point: the primary-source audit found no reviewed concrete-security parameter pin. The paper therefore makes no 128-bit-security, WAN, malicious-security, training-layer, or full-dealerless-Orca claim.

1 Problem overview: Orca's preprocessing and its trusted dealer

1.1 Background: Orca in one page

Orca [1] is a two-party secure computation (2PC) system for neural-network training and inference, engineered around GPUs. Two servers P_0 and P_1 hold additive secret shares of all inputs, weights, and activations; neither learns anything beyond its own shares. Like most practical 2PC systems, Orca splits work into an input-independent *preprocessing* (offline) phase that generates correlated randomness, and a fast *online* phase that consumes it:

- **Linear layers** (convolutions, fully-connected/FC layers) consume *Beaver-triple–style matrix keys*: for a layer multiplying an $M\times K$ matrix by a $K\times N$ matrix, the parties need shares of

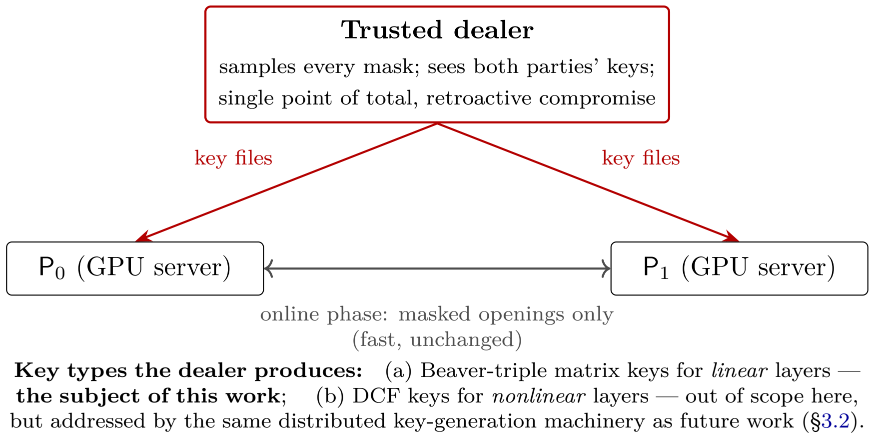

Figure 1: Orca's preprocessing today. The dealer is trusted with every secret of the offline phase. This proposal removes it for the linear-layer keys; the online phase is never modified.

random masks A, B and of the product terms that make the masked multiplication identity work. Online, each party opens masked operands and runs the local computation implemented by `gpuMatmulBeaver`.

- **Nonlinear layers** (ReLU, truncation) consume function secret sharing (FSS) keys — distributed comparison functions (DCFs) — with a different structure.

In stock Orca *both* kinds of keys are produced by a **trusted dealer**: a third machine that samples every mask, computes every correlated value, and ships each party its key files (Figure 1).

1.2 Current status of our pipeline

Over the preceding milestones this project built, and validated on GPUs, a preprocessing pipeline that produces Orca's FC-layer keys from *Ring-LPN pseudorandom correlation generators* rather than from a dealer's coin flips. The pipeline (Figure 2; components defined in §3) already works end-to-end with *real* correlation generators — not idealized stand-ins — at both supported modulus widths, and its output keys are validated by the strongest referee available: Orca's *unchanged* online code reconstructs the correct products (§4.3 gives the measured evidence).

1.3 Historical gaps and current boundary

The original one-process transcript had four dealer/security gaps. The live forward artifact closes the first three at the executable level; the fourth still blocks a deployable security claim. Nonlinear keys remain out of scope.

G1 — Centralized SPFSS setup: closed in the live path. Each process samples only its own sparse noise, participates in distributed GPU-AES DPF key generation, and expands only its own Ring-LPN share. The older one-process transcript retains `build_spfss_keys()` solely as a diagnostic/validation artifact; none of its timings are presented as live protocol measurements.

G2 — Exact share conversion: closed in the hybrid and live path. Batched SCI/IKNP-generated edaBits, daBits, and Boolean triples implement exact $\mathbb{Z}_Q \to \mathbb{Z}_{2^{\mathsf{bw}}}$ conversion. The wrap bit is never opened; the simulator in §3.5 proves privacy at the correlation boundary. The current TCP transport is unauthenticated.

G3 — Randomness separation: closed in the live path. Private roots, masks, and noise come from each process's OpenSSL private DRBG. For every Ring-OLE correlation, the parties exchange full uniform field-coefficient shares whose sum is an exact uniform public polynomial; a short revealed PRG seed is not substituted for this distribution.

G4 — Security parameters: open. The deployed $(c, t) = (2, 8)$ values are feasibility parameters. The exact splittable Ring-LPN distribution and the two-CRT-limb composition have no reviewed concrete classical/quantum advantage bound or accepted parameter pin.

G5 — Scope. Stateful training transitions and nonlinear DCF keys are not implemented by this artifact. The proved/exercised scope is one forward FC matmul; "dealerless Orca" without that qualifier is false.

Contributions. (1) A live, dealer/oracle-free two-process forward-FC preprocessing artifact whose party-local outputs are consumed by unchanged Orca code; (2) an exact algebraic coupling of distributed DPF correction words to standard generation, plus complete role-specific batch simulators; (3) a hybrid-model privacy proof and live realization of exact $\mathbb{Z}_Q \to \mathbb{Z}_{2^{\mathsf{bw}}}$ conversion; (4) a source-to-transcript map and conditional forward-FC theorem with explicit leakage, assumptions, and nonclaims; (5) a reproducible exact-ResNet18-classifier evaluation with ten trials and a matched stock trusted-dealer baseline. The point-DPF and Ring-LPN generator primitives are prior work, not claimed as novel. The systems result is currently a feasibility result: its 3286× median preprocessing/dealer ratio and unpinned Ring-LPN parameters are publication risks, not details hidden behind optimistic language.

2 Motivation: why dealerless, and why on GPUs

2.1 Why remove the dealer

The dealer is the single point of *total, retroactive* compromise: it holds every mask of every computation, so one breach — or one subpoena, or one insider — retroactively breaks the privacy of everything the two parties ever computed. The trust assumption is also operationally awkward: in many two-party deployments (two mutually distrusting companies; a hospital and an ML provider) no mutually trusted third machine exists, and dealer key material for training runs is measured in gigabytes that must be generated, stored, and shipped securely. Removing the dealer replaces an institutional assumption with a cryptographic one.

2.2 Why pseudorandom correlation generators, and which one

A training run consumes correlated randomness for billions of secure multiplications; any protocol paying per-multiplication communication in the offline phase is dead on arrival at that scale. Pseudorandom correlation generators (PCGs) [5, 6] restructure the cost: the parties run one small interactive setup yielding short correlated seeds, then each party *locally* — with zero communication, "silently" — expands its seed into millions of correlated samples.

The atomic correlation we need is *oblivious linear evaluation* (OLE): P_0 holds u, P_1 holds v, and they obtain additive shares of $u \cdot v$. Two OLEs make one Beaver triple; Beaver triples are exactly what Orca's linear-layer keys package. The PCG we build on is the ring-OLE generator of

Boyle et al. [6], which we refer to as the **BCG+20 generator**.[1] In outline, over the negacyclic ring $\mathcal{R}_p = \mathbb{Z}_p[X]/(X^n + 1)$:

1. Each party σ samples c *sparse* secret polynomials $e_1^\sigma, \ldots, e_c^\sigma$ (only t nonzero coefficients each). Its generator input is $x_\sigma = \sum_i a_i e_i^\sigma$ for public random a_i — pseudorandom under the Ring-LPN assumption (§3.8).

2. The target product expands as $x_0 x_1 = \sum_{i,j} a_i a_j \,(e_i^0\, e_j^1)$: all secrecy concentrates in the c^2 products of sparse polynomials, and *sparse times sparse stays sparse* — each product has at most t^2 nonzeros, at positions that are *sums* of the two parties' secret positions with values that are *products* of their secret values.

3. Compact shares of such sparse vectors are exactly what sums of distributed point functions provide (SPFSS, built from DPF trees [3]): kilobyte-sized keys whose full-domain evaluations sum to the hidden sparse vector.

4. Each party locally evaluates its trees everywhere, multiplies by the public $a_i a_j$ (fast NTT polynomial arithmetic), and accumulates. Output: a *ring OLE*, $z_0 + z_1 = x_0\, x_1$ in $\mathcal{R}_p$.

Slot packing — the amortization that makes this affordable. Our primes are chosen so that $X^n + 1$ splits into n linear factors mod p; then the negacyclic number-theoretic transform (NTT) is a ring *isomorphism* $\mathcal{R}_p \to \mathbb{Z}_p^n$, and reading one ring OLE per coordinate ("slot") yields up to $n = 8{,}192$ *independent scalar OLEs* from a single ring-OLE expansion. Because the NTT is linear, each party transforms its own shares locally. In the measured isolated-expansion artifact (§4.3), suite shapes consume $2L\lceil MKN/n \rceil$ ring OLEs (L is the limb count) — e.g. **two** for a 16×32×16 layer at $Q = p_0$, instead of 16,384 per-scalar OLE calls. A self-sustaining factory must reserve next-epoch keygen correlations, reducing usable capacity to at most $n - 3c^2t^2$ and increasing this count (§3.3).

2.3 Why GPUs

The motivation is empirical, not aesthetic. Profiling the expansion (§4.3) shows that **SPFSS full-domain evaluation — millions of independent AES/PRG invocations across thousands of DPF trees — dominates the cost** (∼99% at realistic noise weights; the NTT is under 1%). That workload is embarrassingly parallel and exactly GPU-shaped, and it is why our expansion runs in milliseconds on one GPU (13.3 ms at the smoke parameters; 61–881 ms at t=64 depending on noise regime). The proposed distributed key generation (§3.3) preserves this shape: all trees advance *level-synchronously*, so the depth-14 tree walk has 14 sequential batches independent of how many thousands of trees are in flight. This is not an end-to-end round count: the input adder, correction-word openings, and two-stage payload multiplication add dependency stages to be measured with M1's real transport. Orca itself is GPU-resident; its randomness factory should live on the same hardware, and our keys are written byte-compatibly so Orca's GPU online path runs unmodified.

M, K, N	linear-layer dimensions ($M \times K$ by $K \times N$ matmul)
bw	Orca's ring width; shares live in $\mathbb{Z}_{2^{\mathsf{bw}}}$ ($\mathsf{bw} \leq 32$ here)
n	ring dimension in $\mathcal{R}_p = \mathbb{Z}_p[X]/(X^n + 1)$; smoke artifacts use 8192, preliminary feasibility uses 2^{14}; not pinned
p_0, p_1	current artifact primes $2^{62} - 6 \cdot 2^{24} + 1,\ 2^{62} - 7 \cdot 2^{24} + 1$; not pinned
L, Q	CRT limbs; $Q = p_0 p_1$ (L=1: "q64", $Q = p_0$; L=2: "q128")
d_{DPF}	DPF depth ($\log_2(2n)$ for the uniform-noise artifact)
c, t	Ring-LPN noise parameters: c sparse polynomials, weight t each
λ	target symmetric parameter 128; GPU expansion now uses full 128-bit seeds with separate control-bit PRF outputs

2.4 Notation

3 Live dealerless forward-FC design on GPUs

3.1 Design overview and principles

Figure 2 shows the composed live forward path. Every stage is implemented and gate-verified for the stated feasibility configurations. Components D1–D4 close the earlier centralized-keygen, conversion, randomness, and process-separation boundaries for this scope; D5 remains open. Three principles govern the artifact:

1. **The online path is untouchable.** Keys are written byte-compatibly; Orca's online matmul (`gpuMatmulBeaver`) is both the consumer and the independent validation oracle. (With the integration flag off, baseline Orca is byte-identical — verified.)

2. **Oracle boundaries are explicit.** Every idealized component is marked at the exact line in the source and named in every report; numbers produced with an oracle in the loop are never mixed with protocol-backed numbers in one table column.

3. **Every claim has a mechanical gate.** Components ship with suites that exit non-zero on any failure; one command re-validates the whole chain (§4.2).

3.2 Setting and threat model

Two computationally bounded parties; **static semi-honest** (honest-but-curious) corruption of one party. The conditional theorem assumes authenticated point-to-point channels, AES as a PRG/PRF (λ=128), standard semi-honest security of IKNP OT/Gilboa OLE and BGI DPF, and decisional splittable Ring-LPN for the exact selected sparse-noise distribution (§3.8). The live artifact instantiates AES/IKNP/Gilboa but communicates over unauthenticated loopback, so it is implementation/correctness evidence for the theorem rather than a secure network deployment.

Out of scope: malicious security, external packet attackers, timing and microarchitectural side channels, stateful training transitions, and nonlinear DCF keys. "Dealerless Orca" without the forward-linear/FC qualifier is not claimable.

[1] The construction is specified in Figure 2 of [6]; our codebase and earlier internal reports call the implementation the "Figure-2 engine" after that figure number. The present document uses "BCG+20 generator" to avoid referencing a figure that does not appear in it.

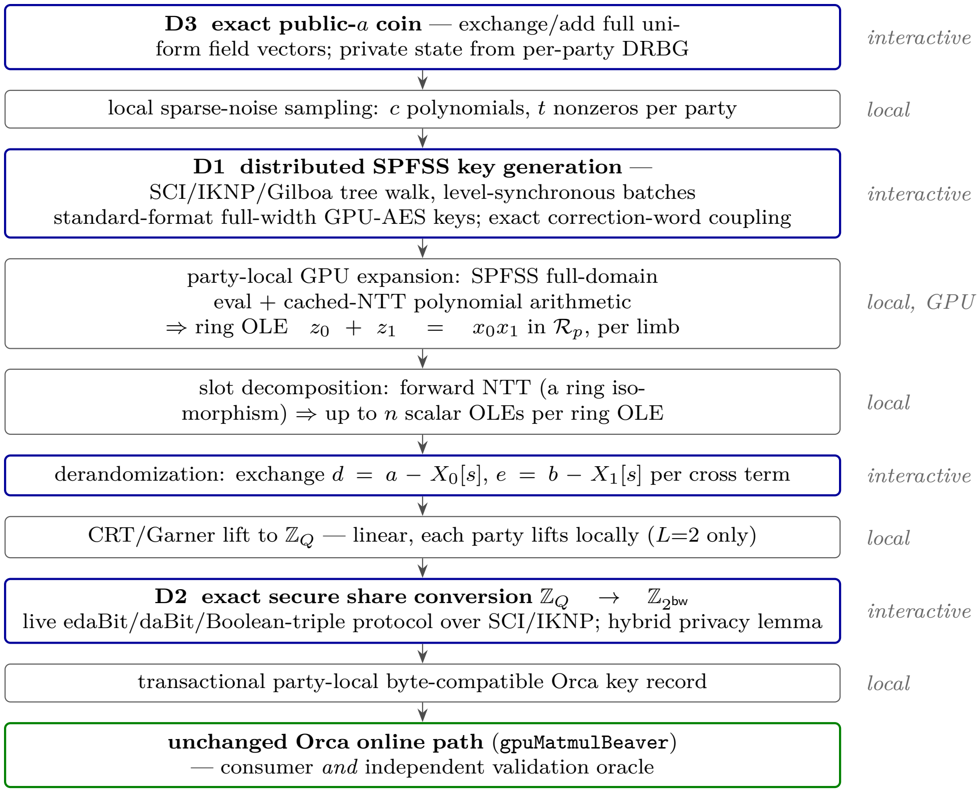

Figure 2: The live forward-FC pipeline; both parties run it as independent processes on distinct GPUs. Blue-framed stages exchange messages over TCP; all other stages are party-local, and Ring-LPN expansion runs on the GPU. The unchanged online consumer runs only in the post-exit checker for this artifact. The transport is loopback and unauthenticated.

Current evidence boundary. The host artifact of §4.3 remains a counted ideal-functionality prototype using the noncryptographic splitmix64 correctness PRG. The live path separately uses full-width GPU AES and real SCI/IKNP/Gilboa transports, exact public-polynomial sampling, and separate process/GPU state. Its proof is conditional on the unpinned Ring-LPN assumption, and its TCP channel lacks authentication; therefore neither artifact establishes an end-to-end 128-bit-security claim.

3.3 D1: Distributed SPFSS key generation (removes G1)

What must be computed. For each pair (k, l) of noise nonzeros, the parties must generate DPF keys for the point function with position $\alpha = \mathrm{pos}^0_k + \mathrm{pos}^1_l$ (a value in $[0, 2n)$: products of degree-$< n$ polynomials have unreduced degree up to $2n - 2$; the negacyclic fold $u[i] - u[i+n]$ is linear and applied locally after expansion) and payload $\beta = \mathrm{val}^0_k \cdot \mathrm{val}^1_l \in \mathbb{Z}_p$. Observe that the required sharing is *already present in the problem*: α is additively shared (each party knows its own summand) and β is multiplicatively shared, with no extra protocol.

Protocol. We adopt the Doerner–shelat two-party DPF key generation [16] in the variant used by the PCG literature [5, 6]. The parties walk the GGM tree level by level. Off the secret path the two parties' nodes are identical, so XORing each party's aggregate PRG expansions leaves the on-path contribution needed for that level's correction word. Selection by the current secret-shared bit of α uses a 1-of-2 OT on λ-bit strings. We keep control/tag bits separate from the full λ-bit seed state, following the formal BGI construction [4]. We do not derive tags from seed LSBs: the S1 target carries 128 secret seed bits and separate tags.

For the arithmetic payload, let A_b be party b's signed leaf-seed aggregate and F_b its signed control-bit aggregate. A first scalar OLE produces $\gamma_0 + \gamma_1 = \beta_0\beta_1 = \beta$. Set

$$d_0 = \gamma_0 - A_0, \quad d_1 = \gamma_1 - A_1, \qquad s_0 = F_0, \quad s_1 = F_1,$$

so $d_0 + d_1 = \beta - A_0 - A_1$ and $s_0 + s_1 = F_0 + F_1 \in \{+1, -1\}$. Two additional directional OLEs share $d_0 s_1$ and $s_0 d_1$; adding those cross-term shares to the local products $d_0 s_0$ and $d_1 s_1$ yields shares $w_0 + w_1 = (d_0 + d_1)(s_0 + s_1)$. The parties open only this standard public key material, $\mathit{finalCW} = w_0 + w_1$; they never open the d_b, s_b, or their sign. The predecessor transcript opened both F_b values. Although their sign is marginally independent of α, conditioning on one party's leaf control-bit vector makes it select the class containing α, so that opening was invalid.

The input bits of α come from a secure ripple adder ($\log(2n) - 1$ Beaver bit triples). This is the concrete integration choice for arithmetic position shares, not a claim of new ripple-adder research. The known 1-OT-per-level walk optimization remains M1 engineering.

Prototype status (implemented; §4.3). The party-separated host implementation generates standard-format DPF keys from shared (α, β) and passes full-domain validation by the *unchanged* existing evaluator on every tree tested: 2,432 key pairs, both CRT primes, depths through $\log(2n) = 14$. It uses ideal-functionality OT, bit triples, and three scalar OLEs, plus the evaluator's non-cryptographic correctness PRG; these are exactly why the executable is not evidence of computational privacy. At depth 14 it records 28 string OTs, 13 bit triples, 3 scalar OLEs, 1,908 logical opened bits, and 3,816 meaningful share bits per tree. Neither bit count is measured transport traffic. The old-sign regression reconstructs the removed sign only in the omniscient harness and confirms that it selects a proper leaf-control-bit class containing the true point; this is a regression model of the old leak, not a privacy proof.

A real-transport path runs the same protocol as two OS processes over SCI/IKNP and Gilboa OLE: 369/369 host-reference keys pass the unchanged evaluator; 88 full-width-AES keys pass batched and per-tree GPU evaluation. The exact coupling lemma below proves that its correction words equal conditioned standard DPF generation, so functional equality is no longer substituted for distribution equality. Single-key privacy still invokes the standard DPF/PRG theorem; this component result does not establish Ring-LPN security.

3.4 S1 security contract for D1 and FC composition

Two-level ideal functionality and state topology. For a public $(\mathit{sid}, \mathit{tree}, d_{\mathrm{DPF}}, p)$, $\mathcal{F}_{\mathrm{DDPF}}$ receives $(\mathit{off}_b, \beta_b, \mathit{root}_b)$ from P_b, where $0 \le \mathit{off}_b < 2^{d_{\mathrm{DPF}}-1}$, $\beta_b \in \mathbb{Z}_p^*$ is canonically encoded, and root_b is that party's uniform 128-bit random-tape draw. It rejects a malformed or duplicate identifier with the same public abort at both parties before consuming correlation. Otherwise it sets $\alpha = \mathit{off}_0 + \mathit{off}_1$ without modular wrap and $\beta = \beta_0\beta_1 \bmod p$, runs standard DPF generation conditioned on the supplied roots, and returns only K_b to P_b. A batch may contain arbitrarily correlated or repeated private inputs, but every tree uses a distinct identifier, independent roots, and fresh primitive correlation.

The stateful $\mathcal{F}_{\mathrm{FC}}$ takes a public ordered topology table whose entries reference public mask-wire handles. A handle names one $\mathbb{Z}_{2^{\mathsf{bw}}}$ mask and its additive shares; repeated handles intentionally reuse those shares. A training-layer row names $(X, W, Y_{\mathrm{bias}}, V_W, V_Y, G)$, its public truncation/optimizer parameters, and whether momentum is enabled; current `FCLayer` requires bias. Each matrix product creates a fresh product-output handle R_C (source-level `mask_Z`) and splits

$$C_0 + C_1 = AB + R_C \pmod{2^{\mathsf{bw}}}.$$

Forward reads (X, W) and emits $X_b \| W_b \| C_{\mathrm{f},b}$; weight gradient reads the same X plus the incoming gradient handle G and emits $G_b \| C_{\mathrm{dW},b}$; input gradient, when enabled, reads that same G and the same W and emits only $C_{\mathrm{dX},b}$.

The source-level forward transition broadcasts and adds persistent Y_{bias} to $R_{C,\mathrm{f}}$ before truncation; the truncated result is the next activation-mask handle. Backward row-sums G to obtain the bias gradient dY, truncates the dX output, and applies both optimizer transitions $(W, V_W, dW) \mapsto (W', V'_W)$ and $(Y_{\mathrm{bias}}, V_Y, dY) \mapsto (Y'_{\mathrm{bias}}, V'_Y)$. Current Orca initializes $W, Y_{\mathrm{bias}}, V_W, V_Y$ to zero; X comes from the preceding truncated output. These persistent operand/state masks are neither resampled nor assumed independent per invocation. Every OT/DPF/OLE/conversion correlation remains fresh even when a public mask handle is reused. The live artifact implements and checks one complete forward FC matmul. Bias/truncation, backward dW/dX, bias-gradient, and optimizer state transitions require a new stateful extension before any training-layer claim.

All ideal calls carry $(\mathit{sid}, \mathit{invocation}, \mathit{tree}/\mathit{slot}, \mathit{phase}, \mathit{ordinal})$ and abort before output on identifier reuse. $\mathcal{F}_{\mathrm{BT}}$ returns private XOR shares of fresh uniform $(a, b, c = a \wedge b)$; the Beaver openings remain real protocol messages. $\mathcal{F}_{\mathrm{OT}}^{128}$ reveals only the selected sender string, and $\mathcal{F}_{\mathrm{OLE}}^{p}$ samples g_0 uniformly and sets $g_1 = x_0 x_1 - g_0$. The composition additionally uses $\mathcal{F}_{\mathrm{COIN}}^{p}$: each party sends a canonical uniform cn-coefficient field vector and their componentwise sum is the exact uniform public Ring-LPN polynomial. $\mathcal{F}_{\mathrm{RINGOLE}}$ returns private shares (X_b, Z_b) with $Z_0 + Z_1 = X_0 X_1$ in the public ring, and $\mathcal{F}_{\mathrm{CONV}}$ handles canonical CRT shares. For canonical $z_b \in [0, Q)$, this functionality computes

$$S = z_0 + z_1, \quad \mathit{wrap} = [S \geq Q], \quad v = S - \mathit{wrap}\, Q,$$

then returns uniform additive shares of $v \bmod 2^{\mathsf{bw}}$ and never opens wrap. The live composition realizes these interfaces over SCI/IKNP and validates its finalized keys only after both party processes exit.

For canonical additive shares, $a_0 + a_1 < 2^{\mathsf{bw}+1}$ and $b_0 + b_1 < 2^{\mathsf{bw}+1}$, so a length-$K$ integer dot product is strictly below $K\, 2^{2\mathsf{bw}+2}$. The FC functionality accepts only public triples satisfying $K\, 2^{2\mathsf{bw}+2} < Q$: reduction modulo 2^{bw} then gives the intended mask product before any reduction modulo Q. This rules out q62/bw32 entirely, caps q62/bw24 at $K \leq 2^{12} - 1$, and is checked before masks are read; q62/bw32 inadmissibility is independently exercised by the executable negative control in `test_orca_zp_bridge`; the live FC path enforces the same public bound before private-mask sampling and never performs a clear value-dependent test. The deployed Ring-LPN GPU expansion uses four domain-separated AES calls per node: plaintexts 0 and 2 produce full 128-bit child seeds, while plaintexts 1 and 3 produce separate control bits. Host/device parity and 88 two-process GPU-evaluated keys are gated. The centralized benchmark kernel still derives roots from one 64-bit `seed_base` and is not security evidence. The live path instead uses OpenSSL-private-DRBG roots/noise/masks and exchanges full uniform field vectors for the public Ring-LPN polynomial. The exact key-distribution coupling and conversion simulator close the algebraic hybrid obligations used by the conditional forward theorem. Concrete Ring-LPN security parameters and an authenticated deployment remain open; no end-to-end 128-bit claim is made.

Leakage and abort contract. Common leakage consists of protocol/session and party-role identifiers; the parameter set, $L, d_{\mathrm{DPF}}, p, n, c, t$, CRT limbs, noise mode, epoch and batch/tree counts; public layer shapes, layouts, invocation kind, wire-handle topology, slot/access order, serialized field lengths, fixed message lengths and dependency schedule; common DPF correction words and *finalCW*; masked openings ϵ, δ, d, e and the specified conversion openings; the full public Ring-LPN polynomial shares exchanged and their sum; and common accept/abort stage. P_b additionally sees only its own DPF inputs/root draw, retained mask shares, local frontier/noise/output-mask state, sender/receiver OT view, triple/OLE shares, authenticated messages, output DPF keys, and emitted Orca fields. The other inputs, roots, keys, mask shares, hidden leaf-control sign, conversion wrap, unused OT messages, OLE masks, and PCG state remain hidden. Public mask-handle reuse is required topology; reuse of primitive correlation or private random-tape draws is prohibited. External packet observers, timing/microarchitectural side channels, active security, fairness, and availability remain out of scope.

Exact D1 hybrid transcript. The proof target is the $(\mathcal{F}_{\mathrm{BT}}, \mathcal{F}_{\mathrm{OT}}^{128}, \mathcal{F}_{\mathrm{OLE}}^{p})$-hybrid protocol instantiated by the current source. *Phase A*: for each of the $d_{\mathrm{DPF}} - 1$ carry positions, obtain one fresh bit triple, reveal both parties' shares of the two masked one-bit values (ϵ_j, δ_j), apply the Beaver equation, and update XOR-shared carry and point bits. This exposes $2(d_{\mathrm{DPF}} - 1)$ logical bits as $4(d_{\mathrm{DPF}} - 1)$ transmitted shares. The private position summands are the Ring-LPN noise positions, so uniform-mode sums are triangular; regular-mode offsets give the corresponding public-bucket shift, not a uniform modular point.

Phase B: at level i, each party locally expands its live seeds, XOR-aggregates left/right seed and flag children, and defines $Z_b = S_b^{\mathrm{L}} \oplus S_b^{\mathrm{R}}$. Two directional 128-bit OTs share $a_i(Z_0 \oplus Z_1)$. Parties reveal their shares of the standard seed correction word and two flag correction words, then advance locally. These are exactly $130 d_{\mathrm{DPF}}$ logical opened bits and $260 d_{\mathrm{DPF}}$ meaningful share bits.

Phase C: one OLE shares $\gamma_0 + \gamma_1 = \beta_0 \beta_1$. With $d_b = \gamma_b - A_b$ and $s_b = F_b$, two directional OLEs share $d_0 s_1$ and $s_0 d_1$. Each party sets $w_b = d_b s_b + x_b + y_b$ and reveals only w_b, giving the standard $\mathit{finalCW} = w_0 + w_1$. Neither d_b, s_b, nor $s_0 + s_1$ is opened. Thus

$$\begin{aligned}\text{logical opened bits} &= 2(d_{\mathrm{DPF}} - 1) + 130 d_{\mathrm{DPF}} + \lceil \log_2 p \rceil,\\ \text{meaningful share bits} &= 4(d_{\mathrm{DPF}} - 1) + 260 d_{\mathrm{DPF}} + 2\lceil \log_2 p \rceil.\end{aligned}$$

Exact coupling to standard DPF generation. After any fixed common-CW prefix, pair the parties' states at each tree prefix. The standard invariant gives equal seed/tag states off the path of α and opposite tags at its prefix, so all off-path values cancel from the joint XOR aggregates. With $a = a_0 \oplus a_1$ and $Z_b = S_b^{\mathrm{L}} \oplus S_b^{\mathrm{R}}$, substituting the two selected OT outputs into the opened seed-CW shares gives

$$sCW = S_0^{\mathrm{R}} \oplus S_1^{\mathrm{R}} \oplus a(Z_0 \oplus Z_1) = \begin{cases} S_0^{\mathrm{R}} \oplus S_1^{\mathrm{R}}, & a = 0,\\ S_0^{\mathrm{L}} \oplus S_1^{\mathrm{L}}, & a = 1.\end{cases}$$

This is exactly the standard losing-branch seed correction. The flag words are

$$tLCW = T_0^{\mathrm{L}} \oplus T_1^{\mathrm{L}} \oplus a \oplus 1, \qquad tRCW = T_0^{\mathrm{R}} \oplus T_1^{\mathrm{R}} \oplus a,$$

so induction preserves the invariant and matches every standard correction word. At the leaves cancellation gives $A = A_0 + A_1 = \mathrm{convert}(s_{0,\alpha}) - \mathrm{convert}(s_{1,\alpha})$ and $F = F_0 + F_1 = t_{0,\alpha} - t_{1,\alpha} \in \{+1, -1\}$. Phase C reconstructs

$$\mathit{finalCW} = (\beta - A)F,$$

which is $\beta - A$ for tags $(1,0)$ and $A - \beta$ for $(0,1)$, exactly the standard final correction. Thus, conditioned on the supplied roots, both output keys equal standard generation deterministically; independent uniform roots give the same joint distribution. This algebraic coupling is separate from the standard single-key privacy theorem and its PRG assumption.

D1 batch-simulation lemma at the ideal boundary. Fix either corrupt party, its complete arbitrarily correlated input/root vector, public order and identifiers, and its full ideal output-key vector. Condition each key on the same root supplied by that party. The following algorithms generate the complete hybrid view with the real conditional distribution. The exact coupling above identifies those conditioned outputs with standard DPF generation; concrete PRG/transport security is separate.

Corrupt P_0. For each tree, take the root seed and common correction words from K_0 and expand the local frontier. At every carry, sample P_0's three bit-triple shares, compute its actual sent shares of (δ_j, ϵ_j), sample the two reconstructed openings uniformly, and set the honest sent shares to the XOR complements. Apply P_0's real Beaver equation, including the public $\delta_j\epsilon_j$ term, and the carry recurrence. Hidden honest triple shares one-time-pad both openings, so induction from carry zero gives the exact joint carry/point-share view.

At every level, derive P_0's aggregates. In the first OT it is receiver: sample the selected output, uniform under the honest sender's fresh mask. In the second it is sender: sample r_0 and record $(r_0, r_0 \oplus Z_0)$. Compute its seed/flag opening shares with the real equations and set the honest shares to complements of the words fixed by K_0. Finally sample its three OLE outputs (γ_0, x_0, y_0) independently, compute d_0, s_0, w_0, and emit only $w_1 = \mathit{finalCW} - w_0$. This ordering also covers zero intermediate values; it does not infer the hidden sign.

Corrupt P_1. Take the root and common words from K_1. Sample its triple shares and masked openings as above, but apply P_1's Beaver equation without the public $\delta_j\epsilon_j$ term. At each level record its first-OT sender pair $(r_1, r_1 \oplus Z_1)$ and sample its selected second-OT receiver output; derive its own seed/flag shares and complement to the fixed common words. Sample (γ_1, x_1, y_1) from the three uniform P_1 OLE marginals, compute d_1, s_1, w_1, and emit only $w_0 = \mathit{finalCW} - w_1$.

Both simulators enforce a consumed-correlation-ID ledger before every ideal call and proceed tree by tree conditioned on the complete prior state. Therefore repeated/correlated private summands do not invalidate the one-time-pad hybrids. S1 treats local coins as independent consumed random-tape draws and makes the root explicit; S3 must prove a concrete, state-consistent CSPRNG interface. In the end-to-end hybrid, only the honest party's hidden PRG stream may be replaced by ideal random strings; the corrupt party's stream and retained state remain consistent.

Conditional forward-FC theorem. Let χ_{U} sample exactly t distinct positions without replacement and χ_{R} sample one position in each of t public equal buckets; in both cases coefficients are independent uniform elements of $\mathbb{Z}_p^*$. Assume decisional Ring-LPN for the exact selected distribution: for uniform $a_1, \ldots, a_c \in R = \mathbb{Z}_p[X]/(X^n + 1)$ and $e_i \leftarrow \chi$, $(a_1, \ldots, a_c, \sum_i a_i e_i)$ is indistinguishable from replacing the final component by uniform R. Also assume the four-call AES expansion is a PRG, the standard BGI single-key theorem, semi-honest IKNP/Gilboa security, authenticated channels, and the BCG+20 Figure-2 reduction.

Under those assumptions, the live forward protocol realizes the forward-matmul restriction of $\mathcal{F}_{\mathrm{FC}}$ with the leakage above. The simulator sets the honest full-polynomial coin share to the ideal public a minus the corrupt share; applies the complete D1 simulator and exact standard-key coupling; replaces Figure-2 expansion by ideal ring OLE; samples each opposing derandomization opening under its fresh hidden OLE share; applies the exact conversion simulator; and adds the fresh uniform output-mask share before serializing $A_b \| B_b \| C_b$. Every replacement conditions on prior batch state and a unique domain separator. This theorem excludes the live unauthenticated TCP transport, malicious parties, and the unimplemented training-state transitions. The measured $(n, c, t) = (8192, 2, 8)$ point is a feasibility configuration with no concrete-security claim.

ID	Proof obligation	Current status
P-CORR	Keys reconstruct $\beta[x = \alpha]$.	2,432/2,432 executable passes
P-DIST	Couple correction words to standard DPF generation conditioned on both roots.	exact algebraic coupling above
P-KEY	One standard key hides point/payload.	standard DPF/PRG theorem; concrete reduction review open
P-POS	Prove the exact uniform/regular Ring-LPN exponent distribution.	contract fixed; distribution/reduction audit open
P-ADD	Simulate either ripple-adder view with explicit triple/opening equations.	closed in bit-triple hybrid
P-LEVEL	Simulate both OT roles and CW shares conditioned on each full key.	closed in OT hybrid
P-PAYLOAD	Simulate the three-OLE view without the removed sign, including zero intermediates.	closed in OLE hybrid
P-BATCH/P-FRESH	Preserve correlated tree inputs and reject correlation-ID reuse before output.	D1 lemma + live distinct-SID controls
P-RNG/P-PCG	Realize private tapes/exact public a and prove Ring-LPN OLE at the selected distribution.	RNG implemented; PCG reduction/parameters open
P-CONV	Realize exact modulo-Q conversion without revealing wrap.	closed in the edaBit/daBit/triple hybrid; transport assumptions explicit
P-TOPO/P-PROC	Match stateful reuse/serialization and two-process execution.	forward FC closed; training state and authenticated deployment open

Table 1: S1 proof contract. Closed rows are hybrid-transcript lemmas only, not a computational-security claim; open rows are binding later-stage gates.

Bootstrapping (and why it is not circular). Key generation consumes three scalar OLEs per DPF tree; expansion produces n slot OLEs per ring OLE. The factory therefore feeds itself precisely when

$$3c^2t^2 \ < \ n, \tag{1}$$

in which case the protocol runs in epochs — a few output slots of each epoch are reserved as the seed OLEs of the next, as in [6] — and only epoch zero needs an external source (a small fixed batch of OT-based Gilboa multiplications). At the smoke parameters $(c, t, n) = (2, 8, 8192)$, key generation consumes 768 scalar-OLE slots and the output/input surplus is $8192/768 = 10.67\times$. Condition (1) *couples the security audit to this design*: if M5 (§4.4) raises t to 64 at $n = 8192$, then $3c^2t^2 = 49{,}152 > 8{,}192$ and the ring dimension must grow with t — consistent with the $n \geq 2^{16}$ parameter points in the literature. Equation (1) is necessary, not a complete epoch budget. If $C_{\text{key}} = 3c^2t^2$, steady-state application capacity is at most $n - C_{\text{key}}$ slots per expanded ring OLE. The current FC artifact counts $2L\lceil MKN/n \rceil$ ring OLEs for two cross directions and L limbs; a self-sustaining implementation needs at least $2L\lceil MKN/(n - C_{\text{key}}) \rceil$ before conversion correlations, discarded tail slots, safety margin, abort/retry handling, and epoch-zero setup. At the preliminary $(2^{14}, 4, 16)$ point only 4096 of 16384 slots (25%) remain, so the 16×32×16 q64/q128 cases need at least four/eight ring OLEs, twice the present artifact count. D5 must close that full allocation and prove consume-once identifiers; this draft does not.

GPU batching. All trees of a generation batch advance level-synchronously: the depth-14 tree walk has 14 sequential batched AES/PRG expansions and OT selections, independent of tree count. The secure ripple adder, correction-word openings, and two-stage payload multiplication add dependency stages. The current TCP implementation measures $6L + 6$ direction switches per batch, independent of tree count; this implementation counter is not an end-to-end network-round measurement.

Cost model (per ring-OLE limb). Each DPF tree of depth d costs λd correlated-OT instances; Ferret-class silent OT amortizes each instance to roughly 1–8 bits of real traffic (implementation-maturity range; M1 replaces this estimate with measurements). With uniform noise there are c^2t^2 trees of depth $\log(2n) = 14$. With *regular* noise (one nonzero per width-n/t bucket), the t^2 product points of each polynomial pair fall on $2t - 1$ predictable diagonals, so the same c^2t^2 trees shrink to depth $\log(2n/t)$, grouped as $c^2(2t - 1)$ SPFSS instances over the smaller domain.[2]

noise, t (c=2)	DPF trees	depth	OT instances	scalar OLEs	est. silent-OT comm.
uniform, t=8	256	14	458,752	768	0.06–0.46 MB
regular, t=8	256	11	360,448	768	0.05–0.36 MB
uniform, t=64	16,384	14	29,360,128	49,152	3.7–29 MB
regular, t=64	16,384	8	16,777,216	49,152	2.1–17 MB

Table 2: Distributed key-generation cost per ring-OLE limb (derived). $n = 8192$, $\lambda = 128$; depth is $\log_2(2n)$ resp. $\log_2(2n/t)$. Scalar OLEs equal three times the tree count. The silent-OT communication estimate excludes the still-unmeasured real OLE transport; M1 replaces both with measurements. The host prototype of §4.3 measures 2 string OTs per level (twice the table's optimized 1-OT-per-level estimate), depth-1 bit triples, and three scalar OLEs per tree.

3.5 D2: PCG-sourced share conversion (removes G2)

The problem. The pipeline holds additive shares over $\mathbb{Z}_Q$ (prime world, where NTTs exist); Orca needs shares over $\mathbb{Z}_{2^{\mathsf{bw}}}$ (hardware world, where they do not). Reducing both shares mod 2^{bw} is wrong whenever the integer share-sum wrapped past Q — and that *wrap bit* depends on both shares jointly, so computing it requires an actual secure protocol.

The protocol (validated two-process artifact). For canonical $z_b \in [0, Q)$, let $S = z_0 + z_1 < 2Q$ and $\ell = \lceil \log_2 2Q \rceil$ (ℓ=63 at q64, 125 at q128). The edaBit functionality samples uniform $R \in \mathbb{Z}_{2^\ell}$, an independent uniform arithmetic first share, and independent uniform XOR masks for its bits; the daBit functionality analogously shares one random bit over $\mathbb{Z}_2$ and $\mathbb{Z}_{2^{\mathsf{bw}}}$. The live artifact realizes one exact daBit with one 128-bit OT, composes one exact edaBit from ℓ daBits (the coefficient-one arithmetic share makes the sum uniform), and realizes each Boolean triple with two one-bit OTs. Under fresh correlations: (1) mask and open $A = (z_0 + z_1 + R) \bmod 2^\ell$; (2) a ripple adder recovers Boolean shares of $S = A - R$; (3) a second adder obtains $w = [S \geq Q]$ from the carry of $S + (2^\ell - Q)$; (4) the daBit converts w to arithmetic shares; and (5) each party computes $r_i = (z_i - Qw_i) \bmod 2^{\mathsf{bw}}$. Thus $r_0 + r_1 = ((z_0 + z_1) \bmod Q) \bmod 2^{\mathsf{bw}}$ and w is never opened.

Independent processes over TCP pass 76 conversions in each of four q64/q128 and bit-width configurations: exact boundary sums $0, Q - 1, Q, 2Q - 2$, random inputs, forced wraps, and layer-shaped inner products, with zero mismatches and a corrupted-output control. The ripple circuit

[2] The 2026-06-10 draft priced the regular-noise row per SPFSS *instance* ($c^2(2t - 1)$ units of depth 14), giving 107,520 OT instances at t=8; the counts below are per DPF *tree*, matching `build_spfss_keys()`, and supersede it. Regular noise's decisive advantage is expansion time and key size (measured 14× and 9.04→5.53 MB at t=64), not OT volume.

consumes $2(\ell-1)$ AND triples, $2\ell-1$ post-mask dependency stages, $5\ell-3$ logical opened bits, and $10\ell-6$ meaningful share bits. The last count excludes byte padding, framing, setup, and OT payloads; the artifact separately measures setup, correlation, and online bytes and direction switches. Dependency stages are semantic depth, not measured network rounds.

Semi-honest privacy at the correlation boundary. A is uniform because R is uniform. Every ripple-AND opening is one-time padded by a fresh Boolean triple, and the B2A opening is $w \oplus d$ for a fresh uniform daBit d; hence these opened values are simulatable and reveal neither S nor w. The output-share distribution also matches the ideal functionality: Q is odd, so multiplication by Q permutes $\mathbb{Z}_{2^{\mathsf{bw}}}$; a uniform arithmetic daBit share makes each $z_i - Qw_i$ uniform subject to the required sum. Given an ideal local output, the simulator solves uniquely for that local daBit share using $Q^{-1} \bmod 2^{\mathsf{bw}}$, samples all local Boolean/edaBit/triple shares with their real marginals, chooses every masked opening uniformly, and completes only the honest transmitted shares. This closes P-CONV in the $(\mathcal{F}_{\text{edaBit}}, \mathcal{F}_{\text{daBit}}, \mathcal{F}_{\text{BT}})$ hybrid. The concrete path remains conditional on SCI/IKNP's semi-honest OT security and an authenticated channel.

What remains for production D2. Two things. *Sourcing:* AND triples come from a binary PCG (the $\mathbb{Z}_2$ instantiation of our own pipeline, or expand-accumulate codes [18]; Ferret [17] yields them at ∼1 bit amortized), edaBits from random bit-shares by the standard construction [19] ($O(\ell)$ AND triples each, one-time batch), daBits from the same batch. *Depth:* the ripple comparator becomes a parallel-prefix (Rabbit-style [20]) comparison — carries obey an associative generate/propagate rule, so all of them come out of a $O(\log \ell)$-depth tree. That depth bounds the eventual interactive schedule, but actual network rounds must be measured after the prefix circuit is implemented and batched.

per $\mathbb{Z}_{2^{\mathsf{bw}}}$ output	q64 (ℓ=63)	q128 (ℓ=125)
AND triples = $2(\ell-1)$	124	248
edaBit bits = ℓ	63	125
daBits	1	1
logical opened bits = $5\ell-3$	312	622
meaningful share bits = $10\ell-6$	624	1,244
post-mask dependency stages, ripple = $2\ell-1$	125	249
post-mask dependency stages, prefix target	∼7	∼8

Table 3: Validated two-process conversion cost per output with executable transcript accounting, plus the prefix-comparator dependency-depth target. The earlier 375/747 column mixed the two shares of the masked-A opening with logical AND/B2A openings; it was not a coherent metric. "Meaningful share bits" is the sum of protocol-defined share widths, not encoded traffic. Measured byte and direction-switch columns are separate; a prefix comparator changes correlation counts and must report its own split.

3.6 D3: Exact public coins and private randomness (removes G3)

The live protocol uses two distinct randomness classes. Private DPF roots, masks, and sparse noise are drawn from each process's OpenSSL private DRBG. For the Ring-LPN public polynomial a, each party samples a full canonical uniform coefficient vector and exchanges it; componentwise addition modulo p is exactly uniform if either contribution is uniform. This costs real traffic, but preserves the required distribution. Revealing one short PRG seed would instead restrict a to the

PRG image and is not substituted without a separate reduction. Session ID, direction, batch, and CRT limb domain-separate all invocations.

3.7 D4: Two-process deployment (realized for forward FC)

The composed artifact runs the parties as two OS processes on distinct GPUs over two SCI-style TCP channels. Each reads only its own sampled noise and emits only its own key record. The post-exit checker reads both records, reconstructs the clear masks only for validation, and invokes unchanged `readGPUMatmulKey/gpuMatmulBeaver`. The runner rejects mismatched preflight manifests, stale output, malformed or swapped records, and unilateral publication, and removes raw party records after validation.

Measured application traffic includes DPF OT/OLE setup, full public-polynomial exchange, cross-term derandomization openings, conversion correlation and online messages, and final synchronization. It excludes TCP/IP framing and base-OT setup. The current runner uses local loopback and the transport is not authenticated.

3.8 D5: Security parameter audit (removes G4)

The assumption behind step 1 of §2.2 is **reducible Ring-LPN** over each deployed prime-field limb: $(a, \sum_i a_i e_i + e_0)$ is pseudorandom for regular sparse secrets e_i. The q128 path is two independently seeded instances over the 62-bit primes p_0, p_1, followed by CRT; it is not one instance over a 124-bit field. The splitting of $X^n + 1$ buys slot packing but exposes reductions modulo sparse low-degree factors. This is the BCG+20 reducible-ring setting, not directly the quasi-abelian group-algebra setting of Bombar et al. [21].

Parameter names matter: BCG+20's total weight is $w = ct$, so its Table-1 row $(c, w) = (4, 64)$ maps to this implementation's per-polynomial $t = 16$, not $t = 64$. The corrected full version does not provide one reproducible projection rule: Section 8.2 derives $cd(1 - (1 - 1/d)^t)$, whereas Section 9.1 uses $w - cd + (c(d-1) + w)(1 - 1/d)^{t-1}$; its literal smallest-factor test selects degree 16 for this row, while Table 1 reports degree 128.

A source audit also invalidated several preliminary estimator outputs. The accepted EUROCRYPT 2024 aggregate calls attack formulas containing $\binom{N-k}{t'}$ and $\binom{N-k-1}{t'}$, so a projected tuple must at least satisfy $t' \le N - k - 1 = d - 1$. The artifact silently evaluates out-of-range combinations. Thus the recorded 57.293-bit $(c, t, d) = (4, 16, 16)$, 218.641-bit $(4, 64, 64)$, and 257.023-bit $(4, 34, 64)$ values are invalid function calls, not attack costs. Mechanically defined rows include 135.12 bits at $(4, 16, 64)$, BCG's degree-128 row at 145.85 bits, and 190.53 bits at $(2, 128, 256)$, but these remain hypothetical finite-field regular-noise model outputs rather than Ring-LPN security estimates.

Sparse-factor reduction creates dependent noise summarized only by an expected weight. Neither source proves its mapping to the estimator's global exact or regular distribution, a lower-tail and rounding bound, coefficient-cancellation loss, applicability to the structured projected code, or the distinguishing- advantage composition over both CRT limbs and the PCG hybrids. Therefore D5 has not pinned n, c, t, p_0, p_1, and this proposal makes no 128-bit classical or quantum claim. All dated conservative-pin CSVs are historical failed-rule transcripts. The measured $n = 2^{17}, c = 4, t = 34$ implementation NO-GO remains valid independently: regular noise requires $t \mid n$, and the uniform layout would materialize 7,272,923,136 host slots (at least 123.6 GB in one process). No replacement parameter is pinned.

boundary	status	executable/proof evidence	consequence
G1 centralized keygen	closed for forward FC	live distributed GPU-AES DPF keys in six cases and the model run; exact correction-word coupling	no dealer/oracle in DPF setup
G2 share conversion	closed in hybrid	live SCI/IKNP conversion plus exact simulator; wrap remains hidden	conditional semi-honest privacy
G3 shared/private RNG	closed for live artifact	per-party OpenSSL DRBG and full uniform public-polynomial share exchange	exact required distribution
process/state separation	closed for live artifact	distinct processes, GPUs, input files, output records; post-exit checker	composition is executable
G4 concrete parameters	**open**	source audit rejects prior pins; no reviewed replacement	no security-level claim
G5 nonlinear/training	out of scope	one forward FC matmul only	no full Orca/training claim

Table 4: Current composition status. "Closed" is scoped to the stated live forward artifact; deployment still requires authenticated transport and a reviewed Ring-LPN parameter pin.

3.9 Composition status

3.10 Per-layer cost model (worked example)

For one FC layer (M, K, N) at L limbs, the current isolated-expansion artifact (which does not reserve next-epoch correlations) uses $2L\lceil MKN/n \rceil$ ring OLEs; derandomization traffic is $4MKNL$ field words; and conversions equal MN (each per Table 3). Key-generation material per isolated ring OLE is in Table 2; the online phase is unchanged because keys are byte-compatible. These are not complete self-sustaining epoch costs.

Worked artifact example, 16×32×16 at q64 (L=1, bw=16, smoke n=8192, c=2, t=8): MKN = 8192 cross terms per direction give **2 isolated ring OLEs**; 32,768 opened words are about 256 kB; and $MN = 256$ conversions use 31,744 AND triples plus 256 daBits in the ideal-correlation prototype. Reserving $C_{\text{key}} = 768$ leaves only 7424 application slots per expansion, so even this example needs at least **4 q64 ring OLEs** in steady state, before conversion-PCG and safety reserves. Every measured artifact number is regenerated from suite CSVs; the production epoch total remains an S2 gate.

4 Experiment setup

4.1 Platform and environment

All measurements in this document were produced on one host: 4× NVIDIA RTX 5000 Ada (32,760 MiB each, compute capability 8.9; driver 560.35.03), CUDA 12.6, Linux 6.8, and a 32-core CPU allocation. Builds run on the host. Two-party tests run as separate processes pinned to distinct GPUs through `CUDA_VISIBLE_DEVICES`; current traffic uses a single-host IPv4 loopback `PartyChannel`/SCI `NetIO` path. The live transport is not authenticated and the application-byte counter excludes TCP/IP and base-OT setup. The GPU NTT backend is a cheddar-derived merged-stage kernel family (signed Montgomery arithmetic; q32/q64/q128-CRT; negacyclic) with the Hadamard product adaptively fused into the inverse-NTT load and forward NTTs of fixed public vectors cached across expansion iterations.

4.2 Validation methodology

Three rules, applied to every artifact. (1) *Independent oracles*: fast implementations are validated against separately written references — the GPU NTT against a host reference (`host_polymul_reference`), the generator's internals against host suites (135/57/36 host validation cases), the end-to-end pipeline against Orca's *unchanged* online code reconstructing the correct product. (2) *Party-separation discipline*: transcript artifacts compute each party's view in separated state; any read across the boundary is an explicitly named oracle. (3) *Mechanical gates*: every artifact ships source + build script + run script + CSV/MD/log; suites exit non-zero on failure. The canonical checkpoint revalidates all host and GPU component gates:

```
RUN_GPU_SMOKE=1 REQUIRE_GPU_SMOKE=1 PATH=/usr/local/cuda/bin:$PATH \
          scripts/run_paper_checkpoint_smoke.sh
```

and must print `ALL GATES PASS`. The exact-shape repeated evaluation is separately regenerated by `scripts/run_two_party_fc_model_scale.sh`.

4.3 Verified artifacts and matched baseline

Single-process real-generator diagnostic. The older suite[3] validates the Ring-LPN engine and slot/Orca algebra in nine q64/q128 uniform/regular cases. It uses centralized DPF key generation and clear exact conversion at explicitly named O1/O2 boundaries. Its narrow stage timings are diagnostic only and are not compared with the live end-to-end measurements.

case	ring OLEs	ideal-OLE equiv.	slots used	opened $\mathbb{Z}_p$ words	contract
q64, 16×32×16, bw=16	2	16,384	8,192	32,768	pass
q128, 16×16×16, bw=32	4	8,192	4,096	32,768	pass

Table 5: Representative rows of the single-process diagnostic (`orca_fc_real_ole_transcript_transcript_suite.csv`). "Ideal-OLE equiv." is the number of per-scalar OLE calls the same layer consumed before slot packing.

Distributed key-generation protocol-logic prototype (corrected 2026-07-29). The artifact[4] implements the two-party key-generation protocol of §3.3 party-separated on the host. Cross-party values flow only through counted ideal functionalities (1-of-2 string OT, Beaver bit triples, and three scalar OLEs) or explicit openings. It emits the standard existing `spfss_host::DPFKey` format, consumed by the untouched `spfss_host::dpfEvalAll`. That evaluator uses a splitmix64 correctness PRG explicitly labelled non-cryptographic; the artifact is therefore a protocol-logic and compatibility prototype, not a secure/private implementation.

The suite (Table 6) generates 2,432 key pairs across six configurations, tree depths 4–14, and both CRT primes. Every pair full-domain-evaluates to $\beta \cdot [x{=}\alpha]$. In every configuration, the first two trees deterministically cover $\alpha = 0$ and $\alpha = 2^{d_{\mathrm{DPF}}} - 2$, factors 1 and $p-1$, and resulting payloads

[3] Artifact `bench_orca_fc_real_ole_transcript`; per-case results in Table 5's source CSV.

[4] `test_distributed_dpf_keygen`; per-case results in `distributed_dpf_keygen_prototype.csv`.

$p-1$ and 1; the remaining trees use random nonzero payload factors. Eight centralized references pass through the same evaluator; five independent root-seed, seed-CW, left-flag-CW, right-flag-CW, and final-CW corruptions fail; and six invalid point/payload encodings abort before correlation is consumed. The field `old_sign_opening_leak_control=yes` records that the omniscient regression model reconstructed the removed sign and observed a proper party-0 leaf-control-bit class containing the true point. This demonstrates the old transcript's distinguishing power; it is not a privacy proof. Every row also requires ideal-mask-draw accounting and a duplicate-correlation-ID negative control to pass. The gate is wired into the repository's one-command re-validation (§4.2).

prime	depth	trees	pass	string OTs	triples	OLEs	logical open	meaningful share bits
p_0	4	512	512/512	8	3	3	588	1,176
p_0	8	512	512/512	16	7	3	1,116	2,232
p_0	11	384	384/384	22	10	3	1,512	3,024
p_0	14	256	256/256	28	13	3	1,908	3,816
p_1	14	256	256/256	28	13	3	1,908	3,816
p_1	8	512	512/512	16	7	3	1,116	2,232

Table 6: Distributed DPF key-generation host results.

Source: `distributed_dpf_keygen_prototype.csv`. For tree depth d_{DPF}, per-tree counts match $2d_{\mathrm{DPF}}$ string OTs, $d_{\mathrm{DPF}}-1$ bit triples, three scalar OLEs, $2(d_{\mathrm{DPF}}-1)+130d_{\mathrm{DPF}}+\lceil\log_2 p\rceil$ logical opened bits, and $4(d_{\mathrm{DPF}}-1)+260d_{\mathrm{DPF}}+2\lceil\log_2 p\rceil$ meaningful share bits. Every row reports `transcript_accounting=pass`, `ideal_mask_draw_accounting=pass`, and `correlation_reuse_control=pass`. Timings are approximately 2.0 ms per depth-14 tree, single-threaded and including validation; they are run observations, not transport measurements. Table 6 proves functional compatibility, edge correctness, and ideal-primitive accounting, not computational privacy or 128-bit security.

Live two-process ResNet18-classifier-layer preprocessing (2026-08-04). The artifact evaluates the exact inference classifier-layer shape 1×512×1000 from Orca's `cnn.h`, at `bw` = 32 and q128 (two approximately 62-bit CRT limbs, not 128-bit security). This is not a full ResNet18 inference or scale-10 truncation run. Each run launches one process per GPU, samples party-local masks/noise/DPF roots, jointly samples the full uniform public Ring-LPN polynomial, generates both OLE directions, securely converts every output, writes one party-local key record, exits both parties, and only then invokes the unchanged `readGPUMatmulKey/gpuMatmulBeaver` checker.

The comparison is negative but decisive: the prototype preserves Orca's final key size and online consumer, but preprocessing is roughly three orders of magnitude slower than matched trusted-dealer key generation and sends 609 MB of application payload for one classifier layer. It is evidence of dealer/oracle removal and exact composition, not a speedup. The dominant remaining systems target is GPU-batched DPF generation plus silent correlation expansion; WAN performance and a security-parameter point are unmeasured.

Performance anchors. Ring-OLE expansion (n=8192, c=2): 13.3 ms (q64) and 26.8 ms (q128) at the t=8 smoke point; at t=64: 881 ms uniform vs. **61 ms regular** (14×), with pair-key bytes 9,044,160 (9.04 MB) vs. 5,529,408 (5.53 MB). SPFSS full-domain evaluation dominates (∼99%); the NTT is under 1% — measured directly, and confirmed by the fact that a 2–8% NTT-side improvement left expansion time unchanged. The oracle-backed conversion stress artifact is bit-exact on 8,516 conversions per configuration; the live two-process OT-backed artifact passes 76 conversions in each of four q64/q128/bit-width configurations, at the costs of Table 3. Byte-

quantity	measured value
warmup / measured trials	1 / 10 (10/10 pass)
regular feasibility parameters	$n = 8192,\ c = 2,\ t = 8$
Ring-OLE correlations / DPF trees, per party	252 / 64,512
critical-path preprocessing, median	35.735 s
critical-path preprocessing, mean ± sample SD	35.813 ± 0.703 s
application bytes sent, both parties	608,860,824 (580.65 MiB)
public-a coefficient shares, both parties	8,257,536 words
final key payload / serialized record, per party	4,108,096 / 4,108,208 B
matched trusted-dealer keygen, median	10.813 ms
preprocessing/dealer ratio, median of trial ratios	3286×
unchanged two-share online checker, median	1.099 ms

Table 7: Exact-shape live evaluation. Timers are local-host observations. The matched dealer row executes stock `gpuKeygenMatmul` twice, sequentially on the checker GPU, with the same shape/masks/key bytes. The online row is likewise two sequential party calls on one GPU, not network latency. Source: `two_party_fc_model_scale_2026_08_04.csv` and its summary/environment manifests.

compatibility: with the integration flag off, baseline Orca output is byte-identical (verified through the real two-party FC test).

4.4 Status gates and remaining experiments

Closed — component correctness. Host and GPU evaluators, full-width AES DPF semantics, distributed-key transport, Ring-OLE expansion, and exact conversion pass their independent reference and negative-control suites.

Closed — live forward composition. Six small/multi-batch cases and the exact classifier-layer run execute DPF key generation, Ring-LPN expansion, derandomization, conversion, transactional key output, and the unchanged Orca online contract across two processes and GPUs.

Closed conditionally — proof boundary. Exact DPF coupling, role-specific batch simulators, the conversion simulator, and the source-to-transcript map establish the forward theorem in the stated ideal-functionality/standard-primitive hybrid. Independent review must be repeated after source changes.

Open — parameters. A reviewed projection, distribution/tail/structured-code and two-CRT-limb advantage analysis must pin (n, c, t, p_0, p_1) before any security-level claim. All performance rows must then be rerun because the current point is feasibility-only.

Open — performance engineering. GPU-batched distributed DPF generation and silent correlation expansion must replace the current host/IKNP bottleneck. Required reporting includes stage time/bytes, setup costs, dependency rounds, peak host/GPU memory, and repeated exact-shape trials.

Open — deployment. Authenticate the transport and measure LAN/WAN executions on distinct hosts. Loopback results cannot predict WAN latency.

Out of scope. Stateful bias/truncation/backward/optimizer transitions, nonlinear DCF keys, active security, and full-model dealer removal require separate protocols and evaluations.

4.5 Reporting and claims discipline

The current supported statement is: "At unpinned feasibility parameters, a live local-loopback two-process implementation generates party-local forward-FC keys with no dealer/oracle in the execution path; those keys match the stock Orca format and pass its unchanged online computation. Conditional semi-honest security follows from the theorem's explicit DPF, OT/OLE, authenticated-channel, and exact-distribution Ring-LPN assumptions."

The only matched quantitative performance baseline is stock `gpuKeygenMatmul`, run at the identical classifier shape, bit width, masks, output bytes, allocator, synchronization boundary, and GPU. The older single-process Ring-OLE stage timers omit most live work and are not a baseline. Prior dealerless PCG systems expose different correlations, parameter assumptions, and applications; no compatible implementation has been measured under this artifact's contract, so no cross-paper speedup ratio is reported.

Unsupported wording includes "128-bit secure," "faster than the dealer," "WAN-ready," "dealerless full Orca," and any training-layer or nonlinear claim. Every future performance table must identify topology, trust/oracle boundary, coverage, exact parameters, counter scope, warmups/trials, and environment; q64/q128 denote one/two 62-bit arithmetic limbs, not security.

5 Related work

Orca [1] defines the host system: FSS-based 2PC ML on GPUs with dealer preprocessing. Beaver's technique [2] underlies its linear-layer keys. BCG+20 [6] already gives the Ring-LPN OLE/triple generator, fully split slot packing, and a semi-honest distributed setup blueprint using generic 2PC plus the Doerner–shelat distributed DPF [16]. Those are inherited primitives. Programmable DPFs [7] reduce distributed generation to constant rounds for feasible domains. Improved DMPFs at IEEE S&P 2025 [8] optimize the multi-point case and its PCG applications. Stationary Syndrome Decoding [9] instead amortizes a shared noise support across correlated instances and explicitly targets the high setup cost of Ring-LPN Beaver generation under a different assumption. More directly, Agarwal–Raghuraman–Rindal's 2026 fully distributed DMPF [10] targets Ring-LPN/Stationary-LPN PCGs with a semi-honest proof and prototype. SLAMP-FSS [11] is a second 2026 multi-point design using tree PRGs and linear systems. The current per-point host DPF is therefore a compatibility artifact and baseline, not a claimed protocol contribution.

Ferret [17] and Silver [18] are candidate silent OT/VOLE transports. Mixed arithmetic–binary conversion uses edaBits/daBits [19] and Rabbit-style comparison [20]; PCGs directly over $\mathbb{Z}_{2^k}$ [12] may remove that conversion. Any-finite-field PCGs [13] and the fully implemented QA-SD/Walsh–Hadamard design [14] are current prime-field architectural baselines that avoid Ring-LPN's FFT choices. Liu et al. [15] provide the field-size-sensitive and regular-noise attack estimators used in the preliminary D5 audit. Bombar et al. [21] analyze a related quasi-abelian-decoding foundation, but their group-algebra theorem is not substituted for the reducible $X^n + 1$ assumption here.

The active polynomial backend is substantially derived from Cheddar [22]; this audit retains its MIT notice and records the reconstructed source pin and local delta. GPU-NTT's Merge and four-step algorithms are separate work by Özcan, Javeed, and Savaş [23, 24] and are currently an external measured baseline. Neither backend is a contribution of this paper.

6 Conclusion and publication assessment

The concrete result is narrower and stronger than the earlier proposal language: a dealer/oracle-free *forward-FC* preprocessing execution now runs across two processes and GPUs, emits stock-format per-party keys, and passes unchanged Orca online evaluation. The exact classifier experiment passes 10/10 measured trials. It also exposes the real current cost: 35.735 s median preprocessing and 608.9 MB sent, versus 10.813 ms matched trusted-dealer generation. The exact DPF coupling, batch simulators, and conversion simulator support a conditional static-semi-honest theorem; they do not supply a concrete Ring-LPN security level or authenticated deployment.

Is it conference-ready? Not yet. The artifact and proof boundary are strong enough for advisor review and for a reproducible systems research checkpoint, but the current manuscript is not ready for a competitive security/systems submission. Three blockers are decisive: no reviewed concrete-security parameter point, a roughly 3286× median preprocessing gap to the matched dealer, and novelty currently centered on integration of known DPF/Ring-LPN/conversion primitives. A cryptography-paper claim would require a new protocol/reduction or parameter result; a systems-paper claim could instead be earned by a substantially faster, end-to-end implementation with convincing ML-scale coverage.

Evidence needed next. (1) An independently reviewed reduction and concrete parameter pin for the exact regular/splittable distribution and two CRT limbs; (2) GPU-batched DMPF/DPF generation with silent OT/VOLE, accompanied by stage-level time, bytes, dependency rounds, and peak memory; (3) all forward linear layers of at least one real inference model, repeated at the pinned parameters, with matched dealer and closest reproducible dealerless-PCG baselines; (4) two-host authenticated LAN and controlled-WAN measurements; (5) a renewed source/proof audit after optimization and a clear novelty statement against the 2025–2026 DMPF/PCG literature. Training, nonlinear keys, and malicious security should remain explicit follow-on work unless separately implemented and evaluated.

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